

ISOMERISM ASSIGNMENT

Structural + Geometrical + Conformational Isomerism | Tautomerism and Optical Isomerism Excluded

Total questions: 128 Structural: 56 Geometrical: 40 Conformational: 32

STRUCTURAL ISOMERISM

Q1.

How many structural isomers are possible for molecular formula C_4H_9Cl

Q2.

Identify total number of structure of ether possible for $C_4H_{10}O$:

Q3.

Total number of structural isomers of C_9H_{18} containing cyclohexane ring.

Q4.

Structural isomers possible for $C_4H_8Br_2$ are -

Q5.

Number of isomeric forms of C_7H_9N having benzene ring will be -

Q6.

The possible number of alkynes with the formula C_5H_8 is -

Q7.

The total number of structural isomers of the compound $C_2X_2Y_2Z_2$ where x, y and z are monovalent groups are:

Q8.

How many geminal dichloride with different structures are possible for $C_5H_{10}Cl_2$?

Q9.

How many structural isomers are possible for C_5H_{10} (excluding stereoisomers)

Q10.

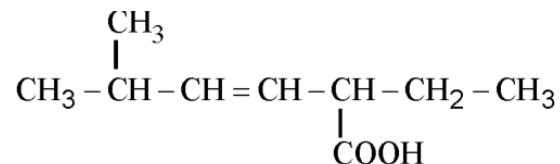
Total number of isomeric alcohols with formula $C_4H_{10}O$ are

Q11.

x = Number of carbon atoms in principal carbon chain

y = locant of methyl group

z = locant of $C = C$



Value of $\frac{x+y+z}{2}$ is ____.

Q12.

How many primary amines are possible for the compounds $C_4H_{11}N$

Q13.

How many open chain structural isomers are possible for C_4H_8 ?

Q14.

How many structurally isomeric dibromo butanes are possible.

Q15.

How many number of all structurally isomeric dienes with molecular formula C_5H_8 are possible?

Q16.

How many aromatic benzene ring containing isomers of formula C_8H_{10} are possible :

Q17.

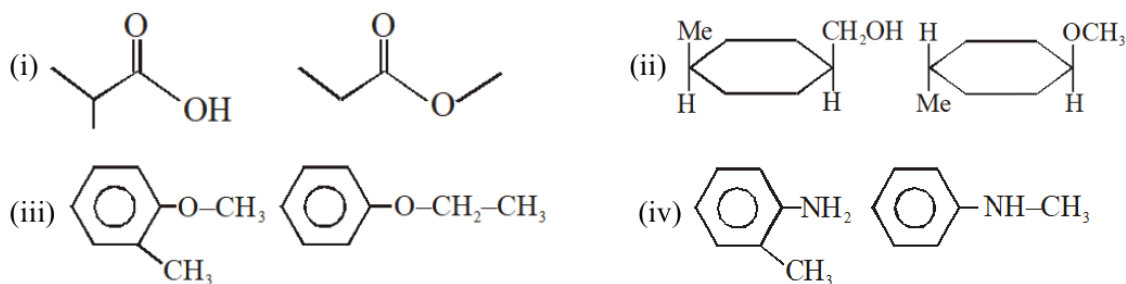
How many tetramethyl benzene are possible :

Q18.

Find out number of structural isomers possible for C_6H_{14} .

Q19.

How many of the following represent pairs of functional group isomers ?



Q20.

The number of structural isomers possible from the molecular formula $C_2H_2FClBrI$.

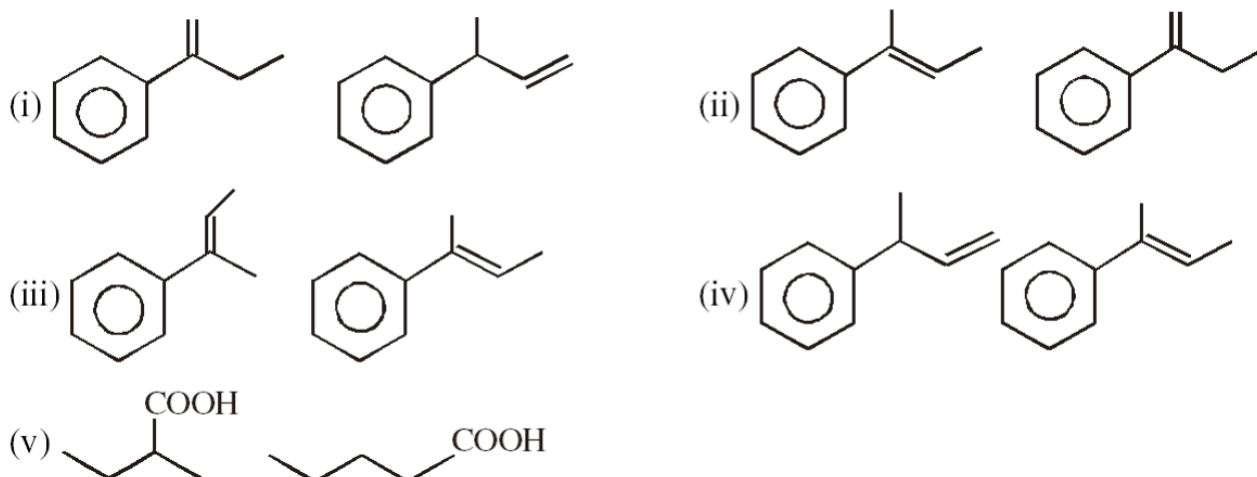
Q21.

Sum of X, Y and Z is :-

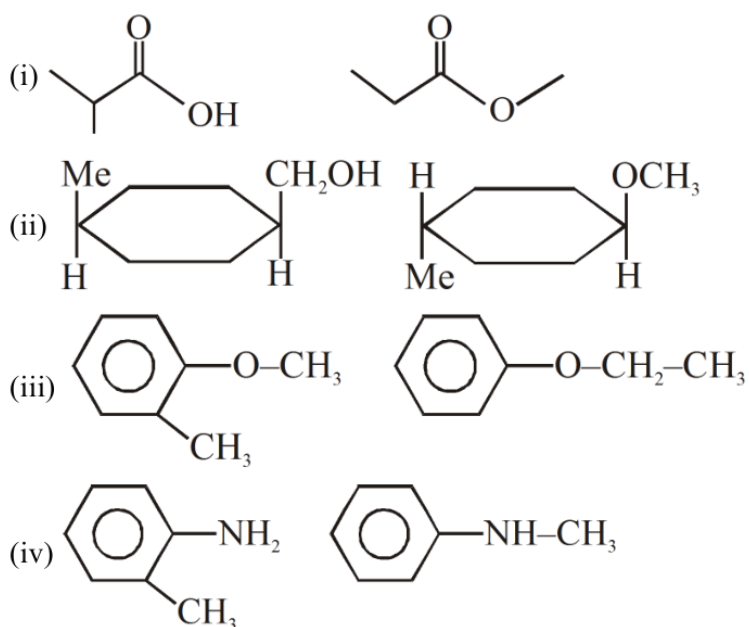
- (a) C_4H_8O No. of aldehydes = X
(b) C_5H_{12} No. of chain isomers = Y
(c) C_4H_8 Maximum no. of alkenes = Z

Q22.

How many among the following parts does not represent positional isomers of each other?

**Q23.**

Which are not functional isomers -

**Q24.**

Calculate the total number of structural isomers of 3°-amines for the molecular formula $C_6H_{15}N$ are?

Q25.

C_7H_9N has how many isomeric forms that contain a benzene ring :-

Q26.

Find the total number of isomers of C_7H_{14} (only 5-membered ring).

Q27.

Possible number of cyclic ketone with molecular formula C_5H_8O is (Only structural isomers)

Q28.

How many structural isomers with benzene ring are possible for molecular formula C_7H_8O :-

Q29.

How many positional isomers are possible for an alkane of molecular mass 72 amu?

Q30.

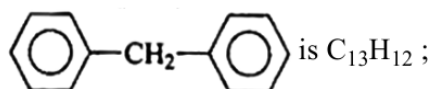
What is minimum molar mass of an alkane which exhibits position isomerism.

Q31.

How many structurally isomeric dibromo butanes (containing parent chain of four Carbon atoms) are possible.

Q32.

The molecular formula of diphenylmethane,



How many structural isomers are possible when one of the hydrogen in replaced by a chlorine atom ?

Q33.

Total number of constitutional isomers (only position isomers) possible for trimethyl cyclohexane is :

Q34.

How many isomers of xylene exist ?

Q35.

How many positional isomers are possible for dimethylcyclohexane?

Q36.

How many position isomers are possible in $C_6H_3ClBr_2$

Q37.

The total number of benzene derivatives having the molecular formula C_7H_8O is -

Q38.

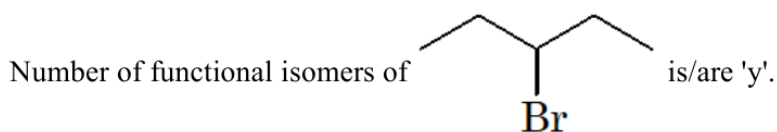
The minimum number of carbon atoms in ether to show metamerism :-

Q39.

The number of metamers of the compound with molecular formula $C_5H_{12}O$ is/are:

Q40.

Number of metamers of $CH_3CH_2-O-CH_2CH_3$ is/are 'x' (including given compound).



Then find the value of $x \times y$.

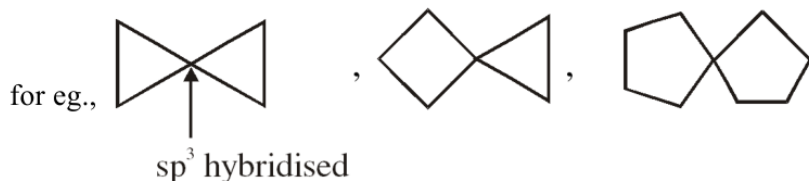
Q41.

Total number of ether structures of $C_5H_{12}O$ are ?

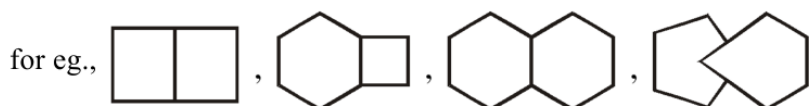
Q42.

Paragraph for Questions 5 and 6

Spiro compounds are those compounds in which two rings share only one carbon atom. Carbon atom which is common to both rings is known as spiro carbon and it must have sp^3 hybridisation.



Bicyclo compounds are those compounds in which two rings share atleast two carbon atoms.



Now answer following :

How many structurally isomeric spiro compounds (having presence of π -bond) are possible corresponding to molecular formula C_7H_{10} in which both rings are three membered?

Q43.

How many structurally isomeric bicyclo compounds (having atleast one **3-membered** ring) are possible corresponding to molecular formula C_6H_{10} ?

Q44.

The number of different chain isomers for C_7H_{16} is _____.

Q45.

How many constitutional (structural) isomers exist for C_6H_6 that have a cyclopropyl ring in their structure.

Q46.

Number of possible cyclic alcohol for $C_5H_{10}O$ is ?

Q47.

If total number of **saturated structural isomers** corresponding to molecular formula C_5H_8 is 'n' then find out the answer of **nth** number question **ONLY** from given below:

- (1) Total number of plane of symmetries in Benzene
- (2) Maximum number of atoms in one plane in benzene
- (3) Total number of geometrical isomers corresponding to $CH_3 - CH = CH - CH = CH_2$
- (4) Total number of plane of symmetries in open book form of H_2O_2

Q48.

The total number of structural isomers for C_6H_{14} is 'x'. The total number of isomeric alcohol (only structural) with formula $C_4H_{10}O$ is 'y'. The total number of aldehydes (structural isomers) with formula $C_5H_{10}O$ is z find $(z + y) - x$.

Q49.

P is the number of structural isomeric amines, with no chiral carbon, which gives isocyanide test and Y is the total number of structural isomeric amines which do not give isocyanide test having molecular formula $C_8H_{11}N$. What is the value of $|P - Y|$?

Note: Every isomer should have phenyl group

Q50.

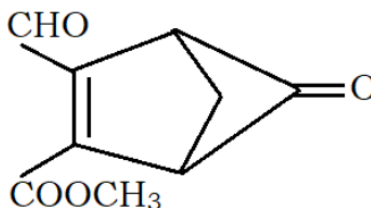
Dicyclo compound possible for $C_5H_8 = x$

Total cyclic structural isomer possible for $C_5H_{10} = y$

Value of $(y - x)^2$ is

Q51.

Degree of unsaturation in the given compound is



Q52.

Analyse the following pairs of compounds.

Write 1, if they are Tautomers.

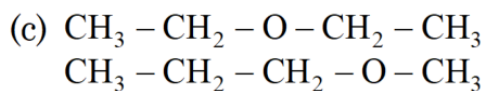
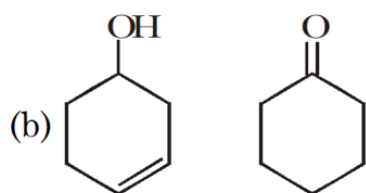
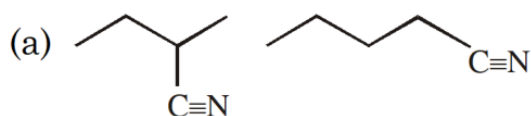
Write 2, if they are Metamers.

Write 3, if they are Position isomers.

Write 4, if they are Functional group isomers.

Write 5, if they are Chain isomers.

Write 6, if they are not isomers.



Write answer of part (a), (b), (c) in the same order and present the three digit number as answer in OMR sheet. For example : If all these answer are 9 then fill 999 in OMR sheet.

Q53.

If the no. of all structural isomer exist for this compound is x (excluding enol) then value of $\frac{4}{5}x$ will be

Q54.

How many of the following compounds will show Ring - chain isomerism

- (i) C_3H_8 (ii) C_4H_8 (iii) C_2H_6
 (iv) C_2H_4 (v) C_3H_6O (vi) C_2H_6
 (vii) C_5H_5N and (viii) C_7H_8O

Q55.

The number of structural isomers possible for $C_6H_6O_3$ containing benzene ring

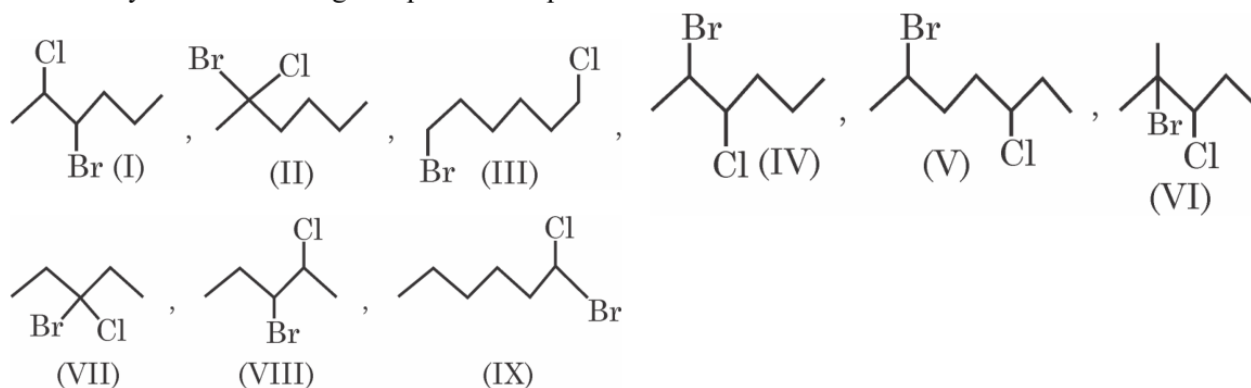
(Do not consider peroxybond containing isomers) is X.

The number of structural isomers possible for C_7H_9N containing benzene ring is Y.

Then $X + Y$ is _____

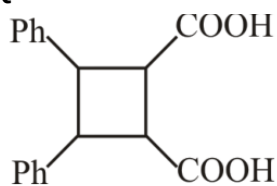
Q56.

How many of the following compounds are positional isomer of 2-bromo-3-chloro hexane?



GEOMETRICAL ISOMERISM

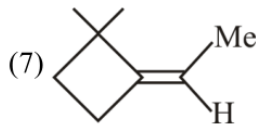
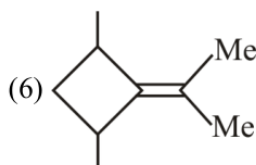
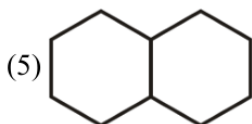
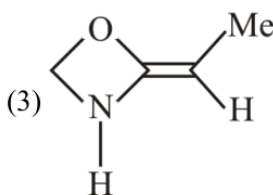
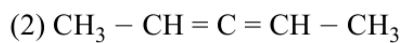
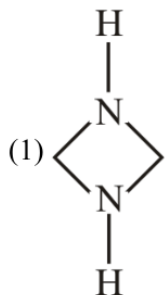
Q57.



Possible geometrical is?

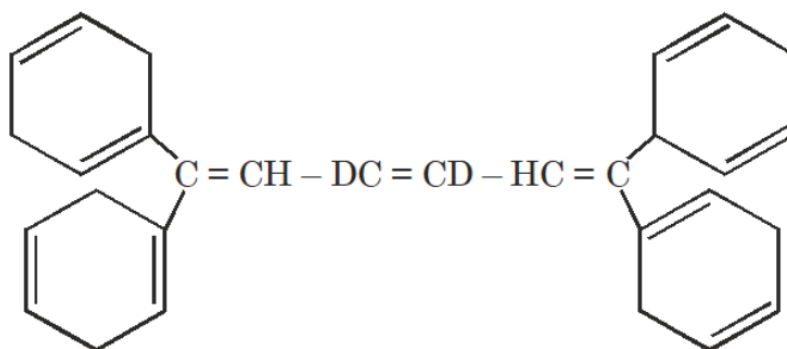
Q58.

Number of compound show geometrical isomerism?



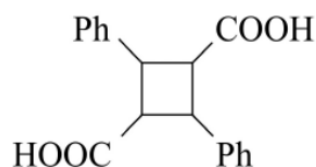
Q59.

Total number of geometrical isomers of the given compound is



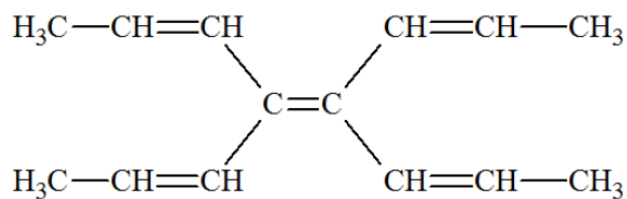
Q60.

How many geometrical isomers are possible for the following compound?



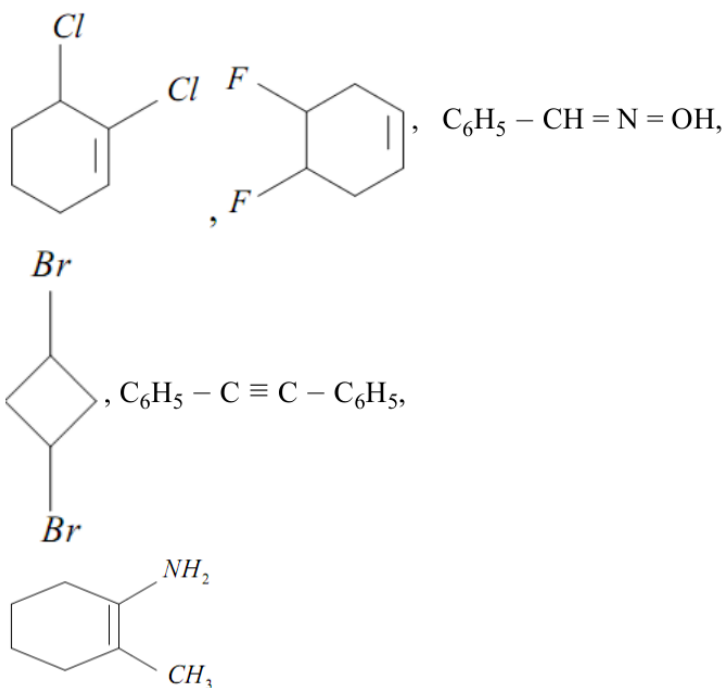
Q61.

How many geometrical isomers are possible for the following,



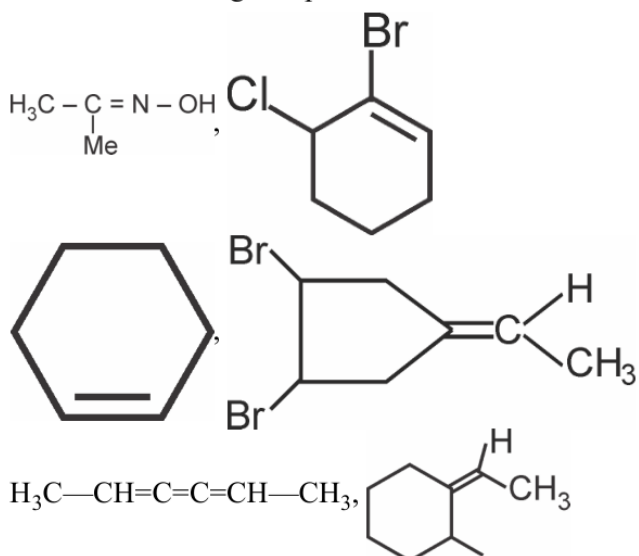
Q62.

Among the following how many compounds can show geometrical isomerism



Q63.

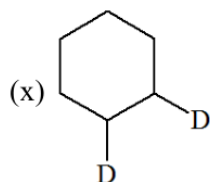
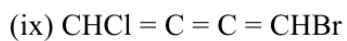
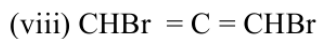
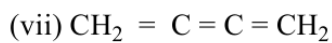
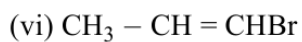
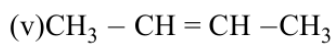
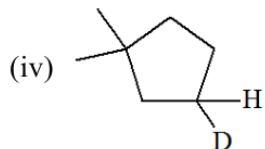
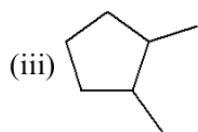
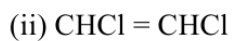
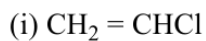
Consider following compounds & write number of compounds which can show geometrical isomerism



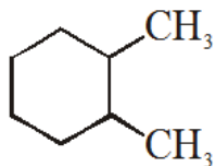
1, 1-dibromoprop-1-ene, Pent-2-ene

Q64.

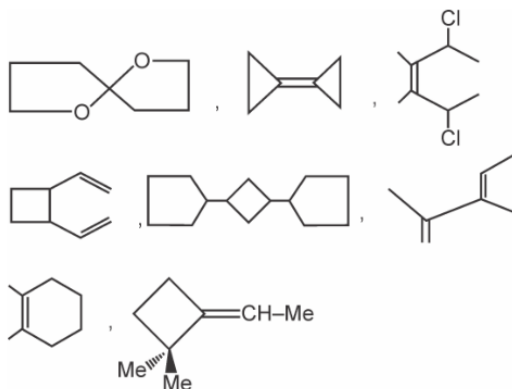
How many of the following can show Geometrical isomerism?

**Q65.**

How many geometrical isomers are possible for:

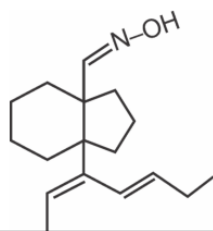
**Q66.**

How many of following compound will show G.I. ?



Q67.

Number of geometrical isomers of compound is :

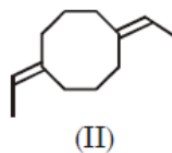
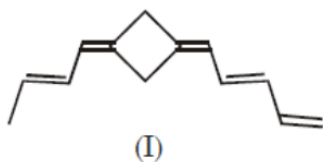


Q68.

How many geometrical isomer possible for :



Q69.



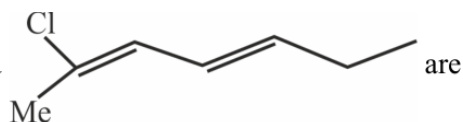
X = Total number of geometrical isomer of (I)

Y = Total number of geometrical isomer of (II)

Then find $(X - Y) = ?$

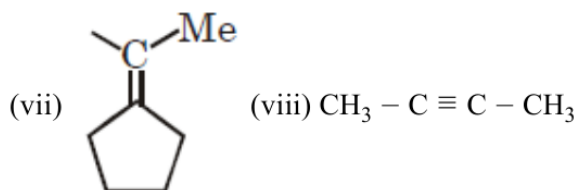
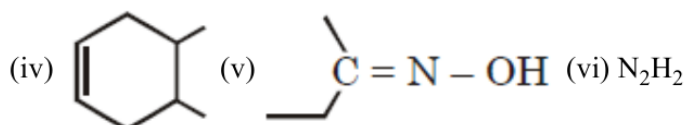
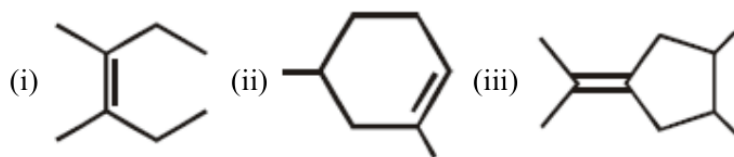
Q70.

Total number of geometrical isomers of compound



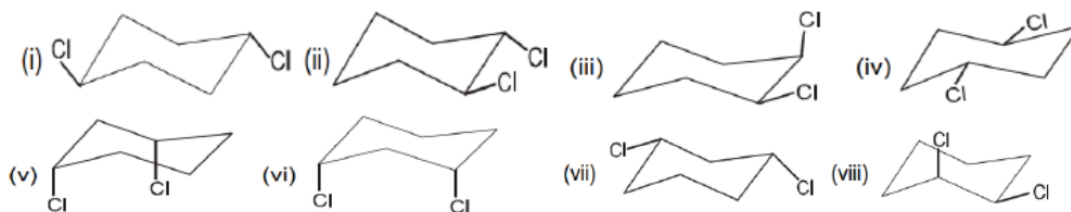
Q71.

How many compound(s) can show G.I.?



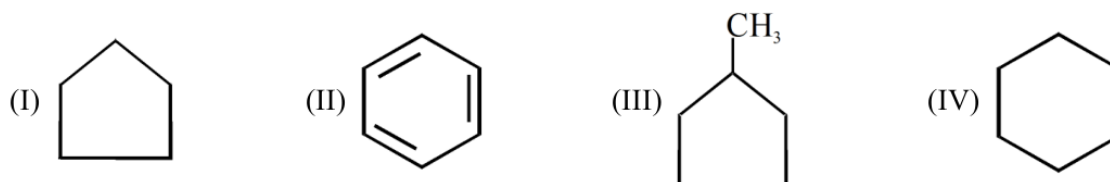
Q72.

How many of the following are cis dichlorocyclohexane.



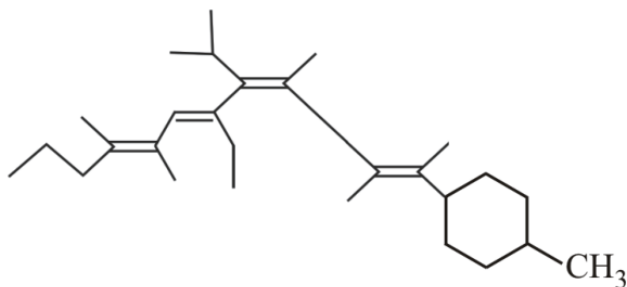
Q73.

Total number of compounds which on dichlorination gives only 3 structural isomers (may not occur by same mechanism in all compounds and even low yield products are to be counted ?



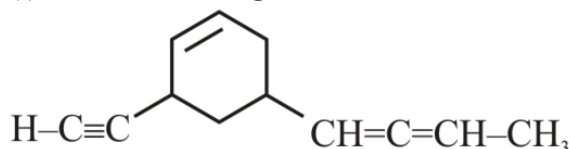
Q74.

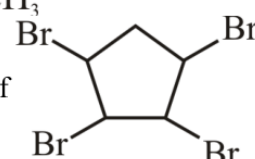
Total number of geometrical isomers of following compound are :



Q75.

(i) Total number of geometrical isomers of the given structure is p.

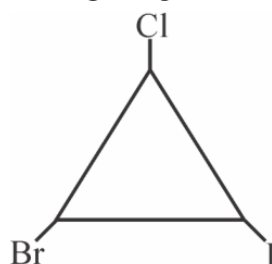


(ii) Total number of geometrical isomers of  is q.

What is the sum of p + q ?

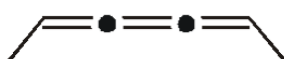
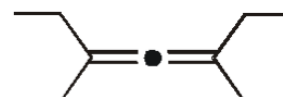
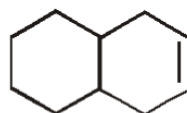
Q76.

Number of geometrical isomers for the following compound is :

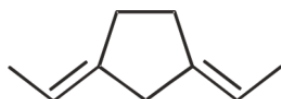


Q77.

Which of the following unsaturated hydrocarbon have hyperconjugative hydrogens & also show geometrical isomerism :



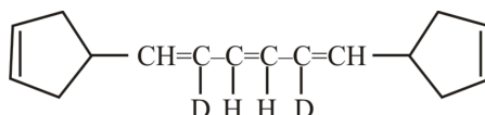
Q78.



Number of G.I. of given compound?

Q79.

Total number of geometrical isomers in the given compound are :

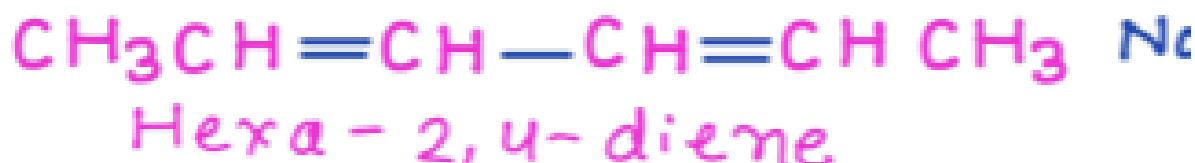


Q80.

In given compounds how many have Z configuration along double bond ?

(i)		(ii)	
(iii)		(iv)	

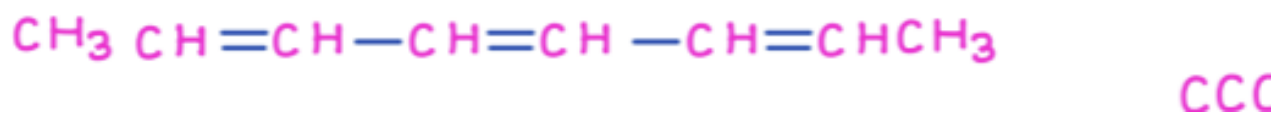
Q81. How many geometrical isomers are possible for the following compound?



Q82. How many geometrical isomers are possible for the following compound?



Q83. How many geometrical isomers are possible for the following compound?

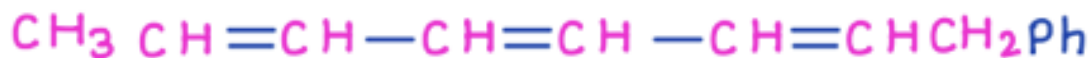


Q84. How many geometrical isomers are possible for the following compound?



$$\text{No. of G.I.} = 2^3$$

Q85. How many geometrical isomers are possible for the following compound?



Q86. How many geometrical isomers are possible for the following compound?

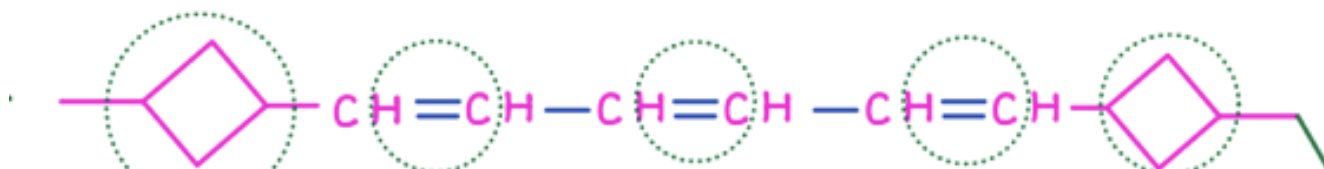


Q87. How many geometrical isomers are possible for the following compound?

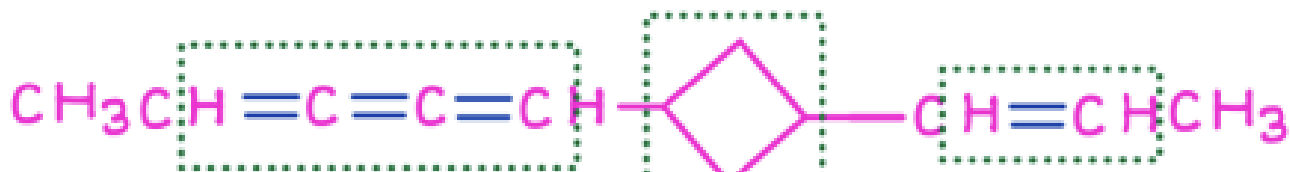
$$= 1$$



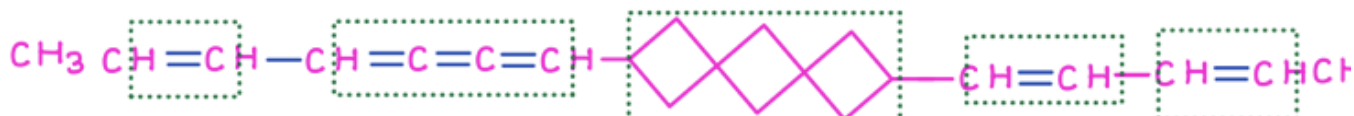
Q88. How many geometrical isomers are possible for the following compound?



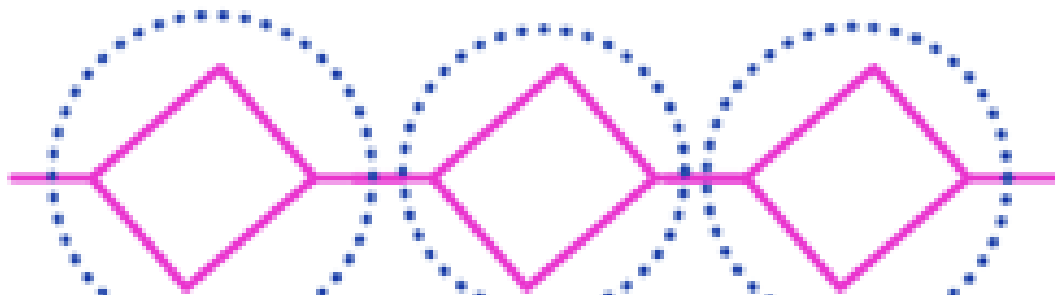
Q89. How many geometrical isomers are possible for the following compound?



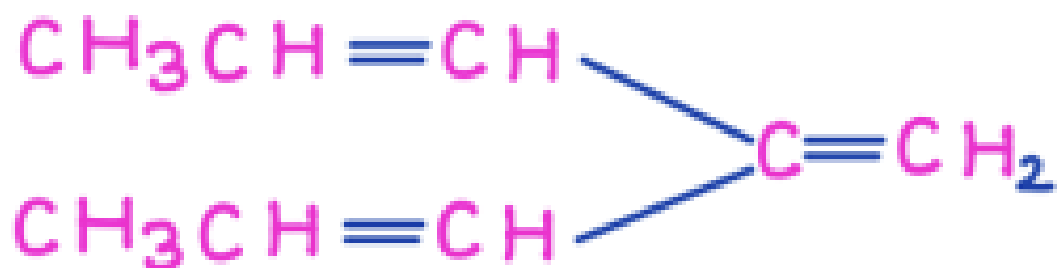
Q90. How many geometrical isomers are possible for the following compound?



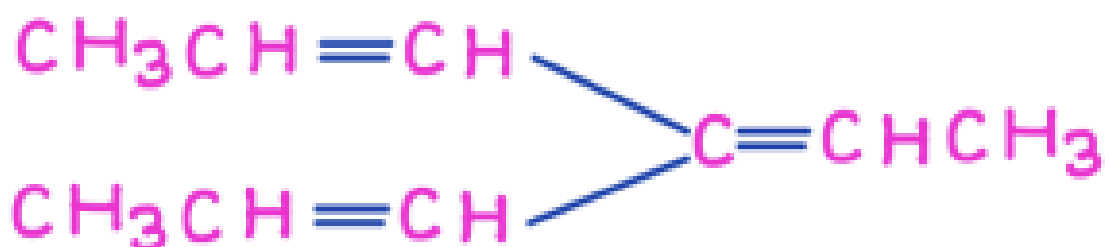
Q91. How many geometrical isomers are possible for the following compound?



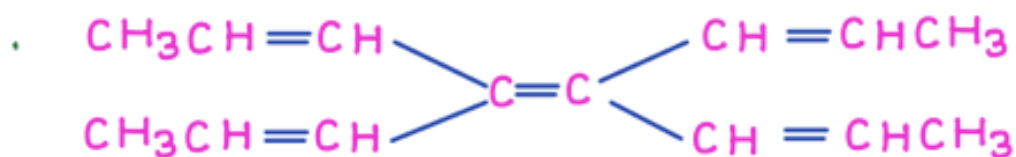
Q92. How many geometrical isomers are possible for the following compound?



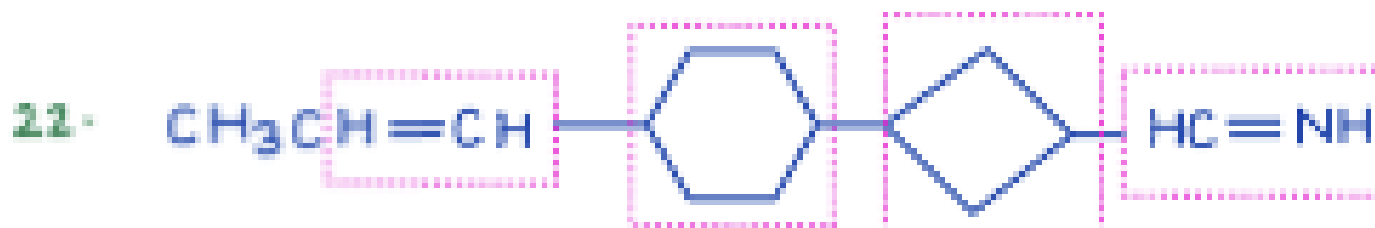
Q93. How many geometrical isomers are possible for the following compound?



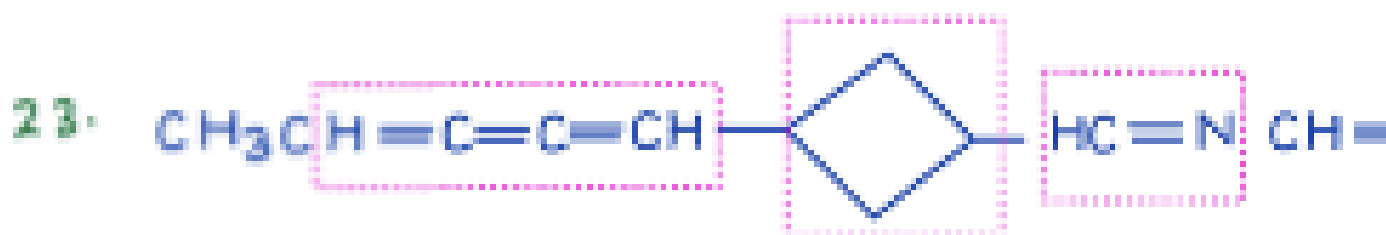
Q94. How many geometrical isomers are possible for the following compound?



Q95. How many geometrical isomers are possible for the following compound?



Q96. How many geometrical isomers are possible for the following compound?



CONFORMATIONAL ISOMERISM

Q97.

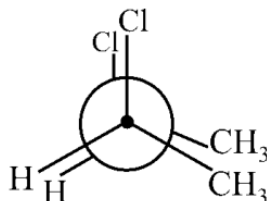
Dipole moment of 1,2-dichloroethane is 1.4 D. Calculate dipole moment of Gauche form, if mole fraction of anti form is 0.8. Dipole moment can be calculated by using formula $\mu = \sum \mu_i x_i$ where x_i represent mole fraction of stable conformation.

Q98.

Total number of stable conformation with non zero dipole moment of 2-bromobutane.

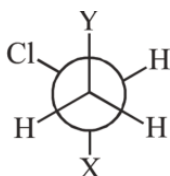
Q99.

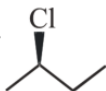
Total number of stable conformation of compound are .



Q100.

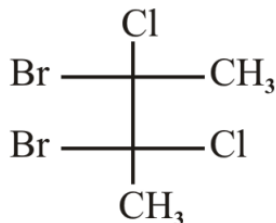
Possible set of (X, Y) so that



is a conformation of  is :

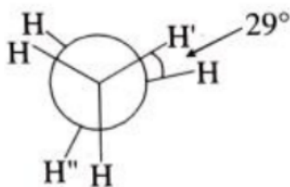
Q101.

The total number(s) of stable conformers with non-zero dipole moment for the following compound is (are)



Q102.

In the following skew conformation of ethane, H'-C-C-H'', dihedral angle is:



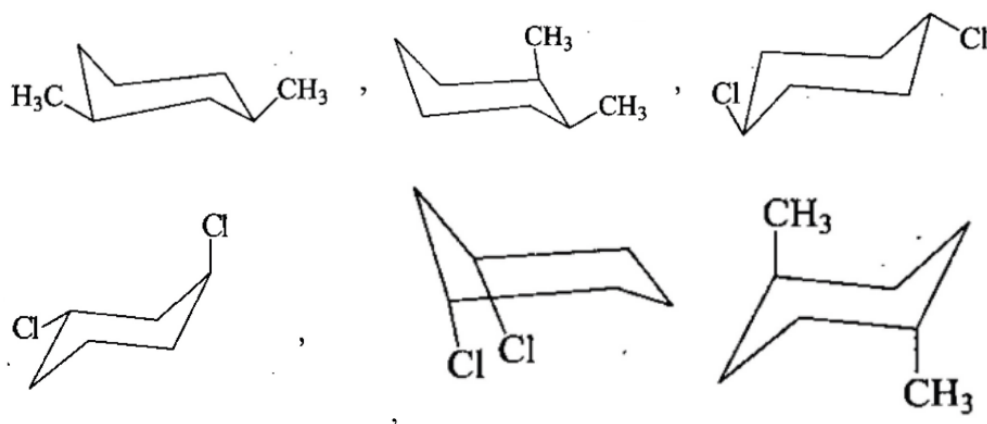
Q103.

For the compound $\text{Cl-CH}_2\text{-CH}_2\text{-Cl}$

if $\mu_{\text{obs}} = 2\text{D}$ and $X_{\text{anti}} = 0.75$ then find μ_{gauche} .

Q104.

Find out number of conformations those have Cis geometry.



Q105.

It is given that for conformational isomers net dipole moment is $\mu_{\text{obs}} = \sum \mu_i \times i$

μ_{obs} = observed dipole moment of the compound

μ_i = dipole moment of stable conformational isomers.

X_i = mole fraction of stable conformer

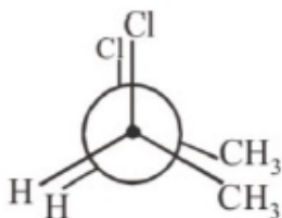
For compound Cl – CH₂ – CH₂ – Cl if $\mu_{\text{obs}} = 0.9$ D

and $\mu_{\text{gauche}} = 5$ D

Find X_{anti}

Q106.

Total number of stable conformation of following compound with non-zero dipole moment are.



Q107.

For the compound A – CH₂ – CH₂ – A if $\mu_{\text{obs}} = 2$ D and $X_{\text{anti}} = 0.75$ then find μ_{gauche} .

(If A = NO₂, μ = dipole moment)

Q108.

Total number of stable conformer having non-zero dipole moment of isopropanol.

Q109.

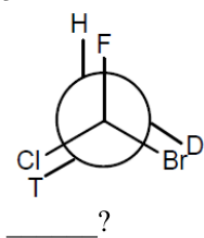
Total number of stable conformer of n-Butane.

Q110.

If dipole moment of 1,2-dibromoethane is 5D. Calculate dipole moment of Gauch form and mole fraction of anti form is 0.6.

Give your answer after multiply with 10.

Q111.



In this structure, F – C – C – T dihedral angle is (15x). The value of (x) in degrees is

Q112.

How many of the following are correct match related to the following,

- (1) → Cis isomer
- (2) → Cis isomer
- (3) → Trans isomer
- (4) → Trans Isomer
- (5) → cis isomer

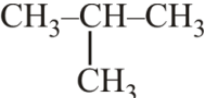
Q113.

How many of the following molecules are most stable in their anti conformation.

- (A) 1, 2-dichloroethane
- (B) 1, 2-difluoroethane
- (C) ethane-1, 2-diol
- (D) 2, 3-dimethylbutane
- (E) 1, 2-diiodoethane
- (F) Hydrazine ($\text{NH}_2\text{-NH}_2$)
- (G) 2-Fluoroethanol
- (H) butane-2, 3-diol (meso)
- (I) butane-2,3-diol (racemic)
- (J) ethane

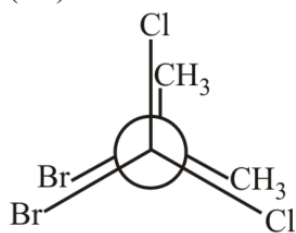
Q114.

Which of the following compounds will exhibit conformational isomerism?

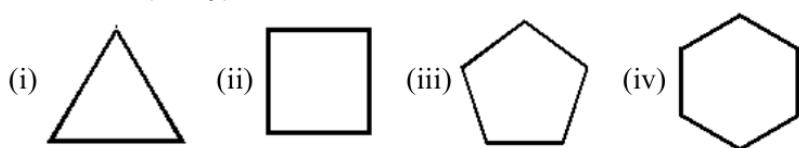
- (i) H_2O (ii) CH_2O (iii) CH_4
- (iv) $\text{CH}_2 = \text{CH}_2$ (v) NH_3
- (vi) CH_3Cl (vii) CH_3OH
- (viii) NH_2OH (ix) CH_3CH_3
- (x) $\text{CH}_2 = \text{C} = \text{CH}_2$ (xi) 
- (xii)  (xiii) $\text{CH}_3\text{-CH}_2\text{-CH}_3$
- (xiv) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$
- (xv) 

Q115.

The total number(s) of stable conformers with non-zero dipole moment for the following compound is (are)

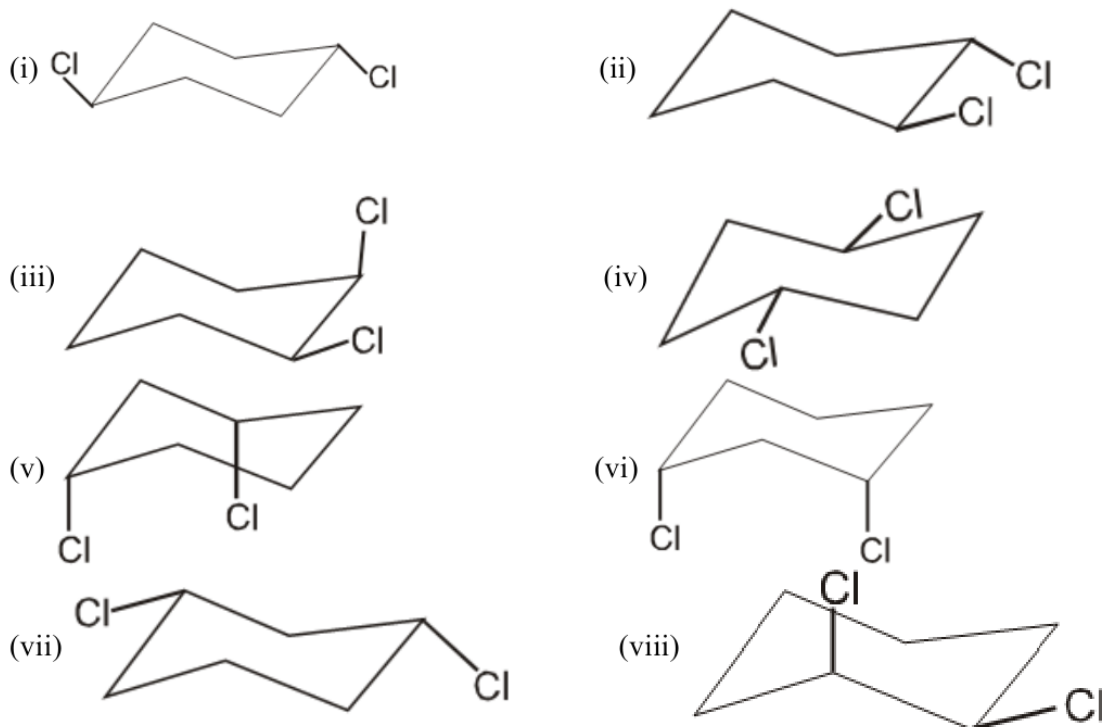
**Q116.**

In how many of the following compounds all carbon atoms are not in one plane in its most stable conformer (if any) ?



Q117.

How many of the following are cis dichlorocyclohexane.

**Q118.**

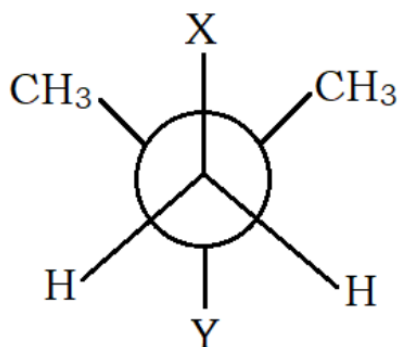
Of all the conformations between C2 and C3 carbon atoms in n-butane, how many are achiral ?

Q119.

$\mu_{\text{obs}} = \sum \mu_i x_i$; where μ_i is the dipole moment of stable conformer and x_i is the mole fraction of that conformer of $Z - \text{CH}_2 - \text{CH}_2 - Z$ in Newmann's projection. If μ_{solution} or $\mu_{\text{net}} = 1\text{D}$ and mole fraction of antiform = 0.82, find μ_{Gauche} .

Q120.

In the Newman projection of 2,2,di methyl butane



How many combination for X and Y respectively correct :

- (i) H and H
- (ii) C_2H_5 and H
- (iii) C_2H_5 and CH_3
- (iv) CH_3 and CH_3
- (v) H and C_2H_5
- (vi) C_2H_5 and C_2H_5

Q121.

In the conformer of cyclohexane

total Axial hydrogen = x

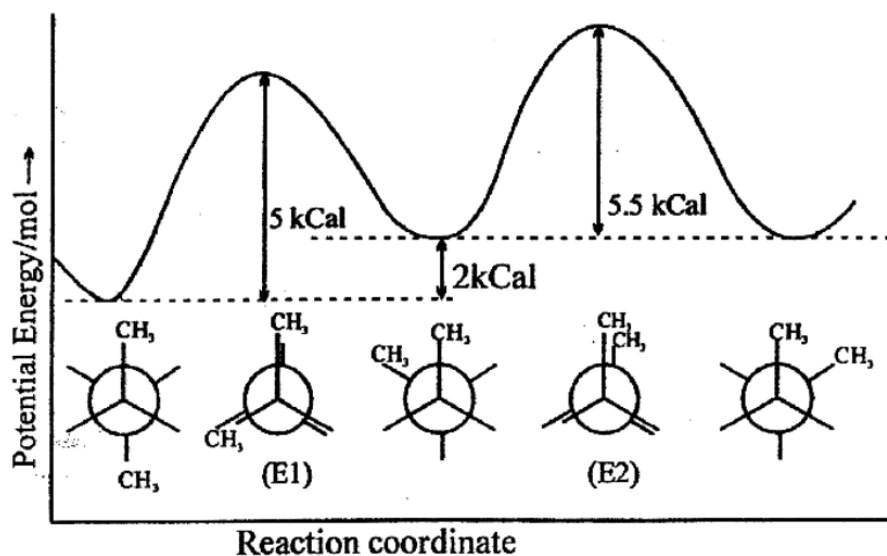
total Number of conformer possible = y

conformer which are stable than chair = z

So write your answer as (xyz).

(If x = 1; y = 1; z = 1 then answer as 111)

Q122.



Butane exist in various coformations in nature like in staggered and in eclipsed form. At any given instant, the probability that a given butane molecule is in anti, gauche, eclipsed-1(E_1) and in eclipsed-2(E_2) conformation is 0.7, 0.2, 0.06 and 0.04 respectively.

If the molar enthalpy of combustion of natural butane is -690 k Cal/mol at 25°C .

Calculate magnitude of Enthalpy of combustion (kcal/mol) of butane if all the butane molecule are in anti conformation.

Q123.

For the conformational isomers, the net dipole moment is $\mu_{\text{obs}} = \sum \mu_i x_i$ where μ_{obs} = observed dipole moment of the compound μ_i = dipole moment of the stable conformational isomers x_i = mole fraction of stable conformers for the compound $\text{Cl} - \text{CH}_2 - \text{CH}_2 - \text{Cl}$, if $X_{\text{anti}} = 0.7$ and $\mu_{\text{obs}} = 2\text{D}$, then if $\mu_{\text{gauche}} = x$ then find $\frac{3x}{10}$.

Q124.

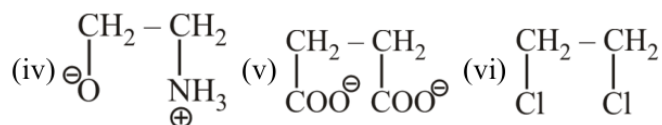
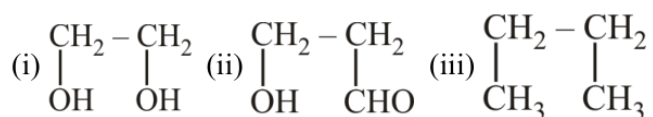
Total number of stable conformer of n-Butane about $\text{C}_2\text{-C}_3$ bond.

Q125.

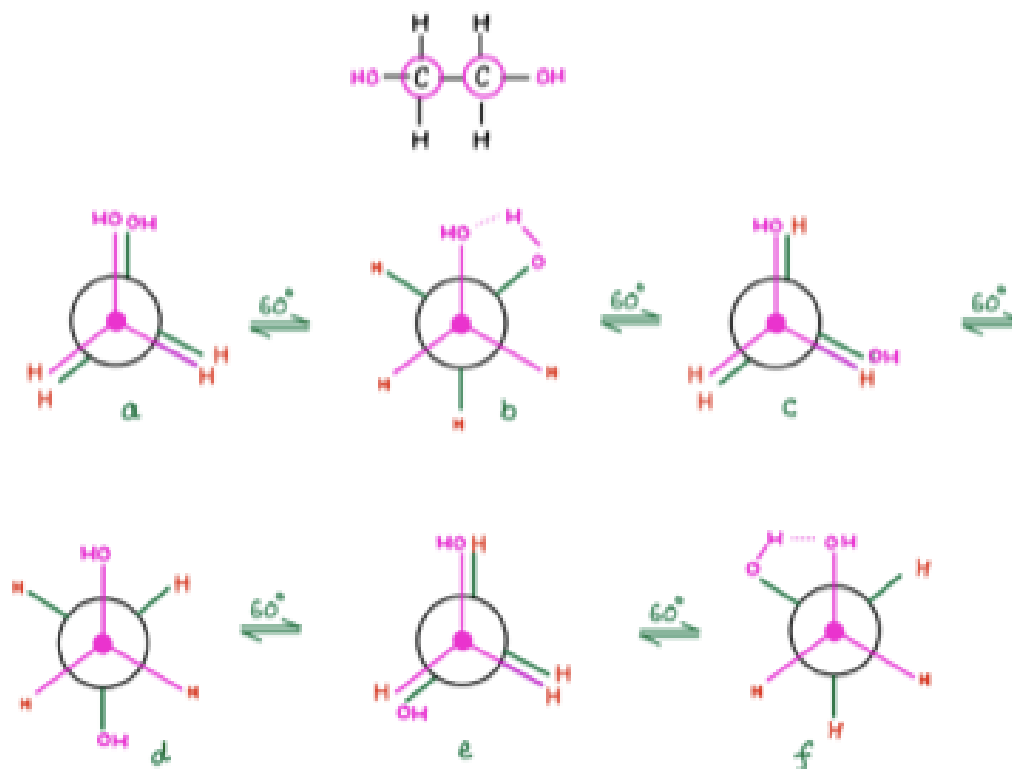
How many statements are correct regarding conformers of ethane is/are : (i) Ethane has infinite number of conformations (ii) The dihedral angle in staggered conformation is 60° (iii) Eclipsed conformation is the most stable conformation. (iv) The conformations of ethane are inter-convertible to one-another.

Q126.

Identify total number of compounds in which Gauche conformation is more stable than it's anti conformation :



Q127. For ethylene glycol, how many of the labelled conformers are tied for the highest stability?



Q128. Using the ethylene-glycol conformers a-f shown in the previous question, write the stability order from most stable to least stable.

ANSWER KEY

STRUCTURAL ISOMERISM

Q1	4	Q2	3	Q3	12	Q4	9
Q5	5	Q6	3	Q7	4	Q8	7
Q9	10	Q10	4	Q11	7	Q12	5
Q13	3	Q14	6	Q15	6	Q16	4
Q17	3	Q18	5	Q19	3	Q20	7
Q21	9	Q22	2	Q23	1	Q24	7
Q25	5	Q26	8	Q27	4	Q28	5
Q29	0	Q30	86	Q31	6	Q32	4
Q33	6	Q34	3	Q35	4	Q36	6
Q37	5	Q38	4	Q39	6	Q40	0
Q41	6	Q42	10	Q43	10	Q44	9
Q45	6	Q46	14	Q47	0	Q48	3
Q49	7	Q50	1	Q51	6	Q52	5, 4, 2
Q53	4	Q54	5	Q55	8	Q56	4

GEOMETRICAL ISOMERISM

Q57	5	Q58	6	Q59	4	Q60	5
Q61	7	Q62	3	Q63	5	Q64	6
Q65	3	Q66	5	Q67	16	Q68	2
Q69	6	Q70	4	Q71	5	Q72	5
Q73	2	Q74	32	Q75	8	Q76	4
Q77	4	Q78	3	Q79	6	Q80	1
Q81	3	Q82	4	Q83	6	Q84	8
Q85	8	Q86	4	Q87	4	Q88	32
Q89	8	Q90	32	Q91	6	Q92	3
Q93	4	Q94	6	Q95	16	Q96	8

CONFORMATIONAL ISOMERISM

Q97	7	Q98	3	Q99	3	Q100	1000
Q101	3	Q102	149	Q103	8	Q104	7
Q105	0.82	Q106	2	Q107	8	Q108	3
Q109	3	Q110	125	Q111	8	Q112	0
Q113	3	Q114	7	Q115	3	Q116	3
Q117	5	Q118	2	Q119	5.55-5.56	Q120	2
Q121	640	Q122	689	Q123	2	Q124	003

Q125	3	Q126	3	Q127	2	Q128	b = f > d > c = e > a
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Note: Questions sourced from the uploaded assignment PDFs retain the original question wording/figures. Additional GI/stability questions are derived only from the user-provided handwritten images.